



ELSEVIER

Solid State Ionics 113–115 (1998) 193–197

**SOLID
STATE
IONICS**

Ion conductivity in amorphous polymer/salt mixtures

D.R. MacFarlane^{a,*}, F. Zhou^a, M. Forsyth^b^a*Department of Chemistry, Monash University, Clayton, Victoria 3168, Australia*^b*Department of Materials Engineering, Monash University, Clayton, Victoria 3168, Australia*

Accepted 31 August 1998

Abstract

Amorphous polymer/salt mixtures based on polyvinyl alcohol and poly(hydroxyethylacrylate) and poly(hydroxyethylmethacrylate) are described. The polyvinylalcohol materials have been prepared by a solvent free hot pressing technique as well as the traditional solvent casting method. The hot pressing technique allows the production of samples which are genuinely free of solvents and thereby has allowed an assessment in this work of the effect of residual solvent on conductivity. The acrylate materials were prepared by direct polymerization of monomer/salt mixtures, thus avoiding the need for solvents. These materials have glass transitions around or well above room temperature, but nonetheless have conductivities as high as 10^{-7} S/cm at room temperature. The temperature and composition dependence of conductivity are also presented. © 1998 Published by Elsevier Science B.V. All rights reserved.

Keywords: Polymer/salt electrolytes; Conductivity

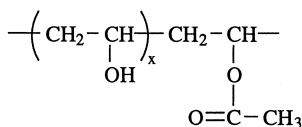
1. Introduction

Amorphous electrolyte materials based on polyvinylalcohol (PVOH) and containing up to 60% by weight of salt have been studied over the last 5 years and shown to have conductivities as high as 10^{-5} S/cm at room temperature [1–3]. Lithium salts such as LiClO_4 and $\text{LiOSO}_2\text{CF}_3$ (lithium triflate) in particular are quite miscible with the polymer. The polymer is often produced by hydrolysis of poly(vinylacetate) and various extents of hydrolysis are available. Previous work [4] has shown that conductivities are typically higher for 88% hydrolysis than they are for 99% hydrolysis. This observation indicates the important, but complex, role played by

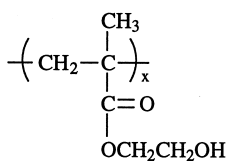
hydrogen bonding in these materials. ^7Li NMR measurements [4] have shown that the lithium ions in these systems remain mobile at and somewhat below room temperature. It has been suggested therefore [4], that it is likely that these materials are mainly lithium ion conductors and are one of the few examples of fast ion conducting amorphous polymer electrolytes. Such fast ion conduction behaviour implies strong decoupling of the conductive modes of motion from the rotational/structural modes of motion that determine the position of the glass transition temperature T_g . Given that it is widely understood that conduction of ions in polyether electrolytes is quite strongly coupled to the polymer backbone motions, it seems likely that the decoupling observed in the PVOH systems may therefore be associated in some way with the hydroxy groups. One of the goals of the work reported here was

*Corresponding author. Tel.: +61-3-9905-4540; fax: +61-3-9905-4597.

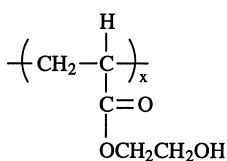
therefore to investigate this conductive behaviour in electrolytes based on other vinyl and acrylic polymers having hydroxy sidegroups, for example poly(hydroxyethylmethacrylate) (PHEMA)–lithium salt and poly(hydroxyethylacrylate) (PHEA)–lithium salt based systems. PHEMA and PHEA can be prepared by standard polymerization techniques from the corresponding monomer.



PVOH - partially hydrolyzed



poly(HEMA)



poly(HEA)

The PVOH materials have previously been made by a solvent casting method and it has always been recognized that residual solvent (dimethyl sulphoxide) remains in the electrolyte even after quite rigorous drying (for example under vacuum, 10^{-4} mmHg, for 72 h). It has been suggested [2] that the residual may be associated in some way with the polymer, however recent evidence [5] suggests that association with the ions may be significant, since the DMSO content tends to increase with salt content. A solvent free route to these electrolytes is thus desirable.

In this work we have prepared polyvinylalcohol and related hydroxy polymer/lithium salt electrolytes by solvent free methods. In the case of the PVOH/salt mixtures the material is prepared by hot pressing a physical mixture of the powdered components. In the case of the acrylate systems, a solution of the salt in the monomer is polymerized directly to a glassy material.

2. Experimental

PVOH/salt systems were prepared by a hot pressing method which involved careful grinding and mixing of the dry components under nitrogen and

then hot pressing the sample at 7 tonnes for 2–4 h at 110°C. Typical sample size prepared in this manner was 14 mm diameter $\times$ 0.1 mm thick. The resultant materials were transparent, but slightly hazy, and visually homogeneous. Examination of the material by optical microscopy revealed no signs of remaining salt crystals, except at the highest salt content, where some regions of crystallinity were observed. After pressing, the samples were stored under dry conditions. Prior to conductivity and DSC measurements, the samples were held under vacuum ($P < 0.1$ torr) for 48 h at room temperature.

Poly(hydroxyethylmethacrylate)–Li triflate and poly(hydroxyethylacrylate)–Li triflate materials were made by dissolving the required amount of lithium triflate in the liquid monomer (all materials from Aldrich). The monomer solution was then polymerized by addition of a small amount of $\text{K}_2\text{S}_2\text{O}_8$ and heating to 90°C. A small amount of tetramethyleneglycol diacrylate was also added as a crosslinker. The resultant polymer/salt materials were hard, clear, homogeneous solids. The salt concentrations studied were chosen to be comparable on a mole(Li):mole(-OH) basis with the PVOH systems.

Differential scanning calorimeter thermograms (Perkin Elmer DSC 7) were obtained over the temperature range -100°C to 100°C . Conductivity was determined by placing a disc of the material into a spring loaded conductivity cell and measuring the impedance spectrum over the range 20 Hz to 1 MHz. Samples were cycled between 0°C and 70°C a number of times before the conductivity was measured as a function of decreasing temperature. This cycling caused a slight increase in the conductivity probably due to the improved sample–electrode contact. The Cole–Cole plot of the impedance data showed one main touch down on the real axis and this was assigned to the ion conduction process in the sample.

3. Results and discussion

The DSC glass transitions were considerably better defined in the case of the hot pressed PVOH samples than is normally the case for solvent cast PVOH electrolytes [2,4]. A typical family of traces is

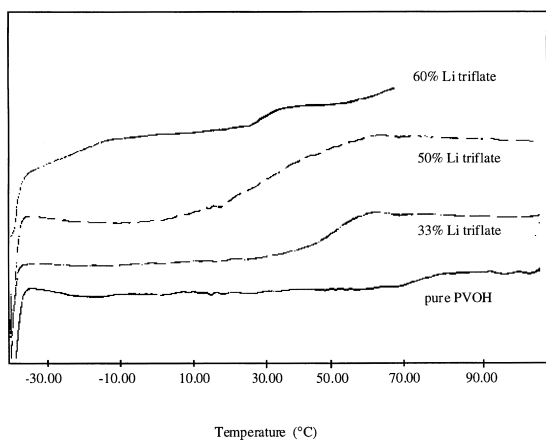


Fig. 1. Typical DSC traces of hot pressed amorphous PVOH electrolytes. The salt contents indicated are weight percent in 88% hydrolyzed PVOH.

shown in Fig. 1, in which the fairly clear glass transition in the pure PVOH and 33% salt samples gives way to a rather diffuse transition in the 50% sample and possibly multiple transitions in the 60% sample. Glass transition temperatures, measured at the mid-point of the transition, are plotted as a function of salt composition in Fig. 2. A clear downward trend is observed as a function of salt content, the presence of the salt appearing to disrupt the hydrogen bonding which dominates the glass transition event in PVOH.

Conductivity of the PVOH/Li triflate samples is plotted as a function of inverse temperature in Fig. 3. The PVOH/Li triflate materials prepared by hot pressing in this work exhibited conductivities which

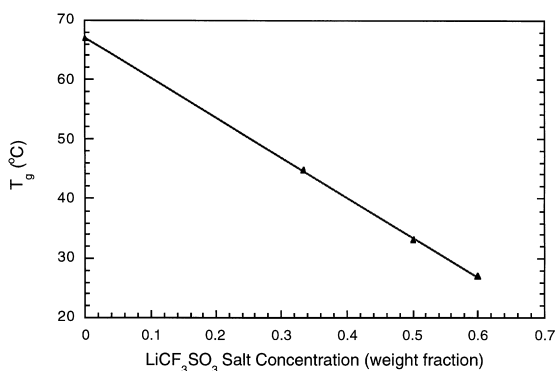


Fig. 2. Mid-point glass transition temperature as a function of salt content for hot pressed PVOH lithium triflate electrolytes.

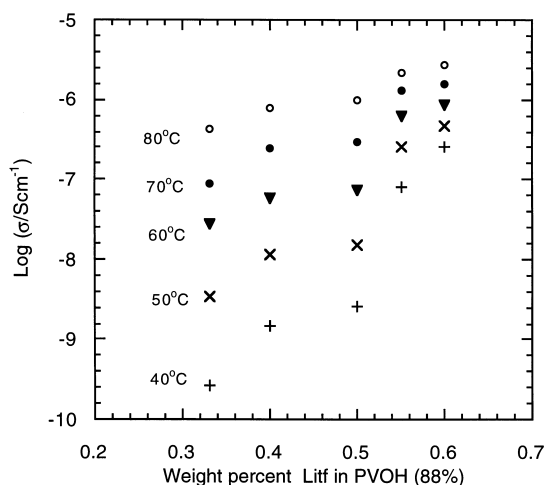


Fig. 3. Conductivity isotherms as a function of $\text{LiOSO}_2\text{CF}_3$ content in PVOH (88% hydrolyzed). All samples prepared by hot pressing.

were consistently one order of magnitude or more below that of the comparable solvent cast composition. For example, for PVOH/Li triflate (1:1 by weight), $\sigma = 10^{-8.5} \text{ S/cm}$ at 40°C , ($T_g = 40^\circ\text{C}$). Since solvent cast PVOH electrolytes are known to contain residual DMSO [2], it is likely that this DMSO is contributing to the enhanced conductivity in the solvent cast samples. Residual DMSO contents have been determined in this work [5] and previously [2] by GC methods. To determine the extent to which this residual DMSO is the source of enhanced conductivity, an amount of DMSO approximately equivalent to the residual was added to the hot pressed compositions at the time of mixing. The conductivity of this sample is compared with the hot pressed and solvent cast samples of equivalent composition in Fig. 4. These samples showed conductivities comparable to the solvent cast material. It is possible, however, that inhomogeneity remains in the hot pressed samples containing no DMSO and that the presence of DMSO causes a more rapid homogenization of the samples during the hot pressing stage. To test this hypothesis a sample was subjected to further heat treatment (70 h at 70°C under vacuum). No significant change in conductivity was observed. In summary, the evidence suggests that the solvent casting route to PVOH electrolytes is susceptible to artificially high conduc-

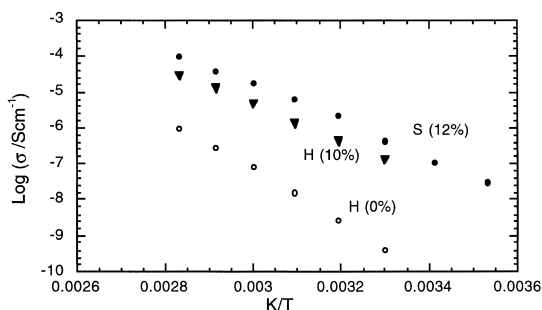


Fig. 4. Comparison of the conductivity as a function of temperature of hot pressed and solvent cast polyvinylalcohol based electrolytes at 50 wt.% $\text{LiOSO}_2\text{CF}_3$. H and S indicate hot pressed and solvent cast samples, respectively, the DMSO content being indicated by the weight percent noted.

tivities as a result of the presence of residual solvent. This solvent may be acting as a simple, classical plasticizer or may be promoting also the ion dissociation step of the conduction process.

These results indicate therefore, that the previous conductivities measured for PVOH-based electrolytes were to some extent enhanced by the plasticizing effect of the DMSO. The underlying conductivity is somewhat lower than previously thought, but nonetheless still substantial at T_g of each system. Despite the possibility of inhomogeneity in the 60% Li triflate sample, it nonetheless exhibits the highest conductivity at all temperatures (Fig. 3). Since T_g falls steadily with increasing salt content, the lower T_g of the 60% salt sample is certainly a factor in its higher conductivity. However, comparing the conductivities of each composition at their respective T_g s shows that this is not the primary influence on conductivity. The composition dependence of conductivity shows a kink in each isotherm at 50% salt content, beyond which the conductivity rises more sharply. This is reminiscent of a percolation type threshold in which connectivity of conducting zones begins to take place. If this was the case it would suggest that the materials become more molten salt-like in terms of their conduction mechanism at the higher salt contents. Connectivity effects have been implicated previously in the conduction mechanism of phase segregated materials [6].

Fig. 5 presents the conductivities of the systems

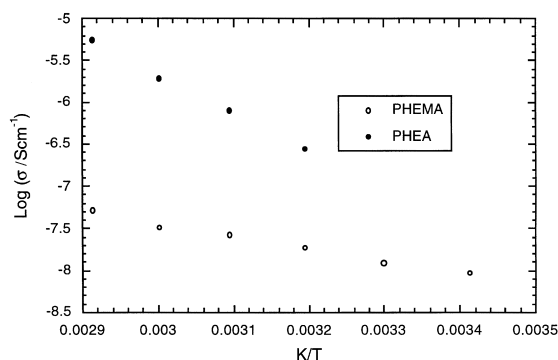
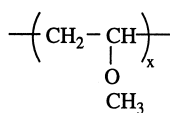


Fig. 5. Conductivity as a function of temperature for poly(hydroxyethylacrylate), (PHEA) and poly(hydroxyethylmethacrylate) (PHEMA)–lithium triflate mixtures.

prepared by direct polymerization of monomer plus salt mixtures. At mole[OH]:mole[Li^+] ratio = 1:0.374, the PHEA and PHEMA-Li triflate systems had conductivities of $10^{-6.5}$ and $10^{-7.7}$ S/cm at 40°C, respectively. This salt concentration is equivalent to the 1 g PVOH:1 g Li triflate composition on a mole ratio basis, though notably it corresponds to a lower salt content on a mass ratio basis. To ensure that the materials were not holding residual monomer or low molecular weight oligomers, samples were subjected to prolonged treatment under vacuum (72 h at 85°C). No change in conductivity was observed. The glass transition temperatures of these samples were not easily identified, however, they appeared to be $\sim 30^\circ\text{C}$ and 40°C , respectively. The main chain methyl group is observed to raise the glass transition of the pure PHEMA polymer as compared to PHEA and this effect still appears to be operative in these salt/polymer mixtures.

It thus appears that the hydroxyacrylate polymers are also able to support high ionic conductivities at and below their respective T_g s; the level of conductivity observed, in particular in the case of the PHEA systems is higher than the corresponding PVOH system. In contrast we have previously shown [4] that similar electrolytes based on poly(vinylmethyl ether), the methyl ether derivative of PVOH, has a very low conductivity at room temperature. This therefore, associates the observation of ion conductivity in the systems described here with the presence of the pendant hydroxy group.



poly(vinylmethyl ether)

4. Conclusions

A new method for the preparation of PVOH/salt electrolytes has been described, which does not involve solvent casting. Samples prepared by this method show significantly less conductivity than solvent cast electrolytes of the same nominal composition. Nonetheless, the PVOH systems remain conductive of ions below their glass transition temperatures. Related polymers which also contain hydroxy groups, but spaced away from the main chain by acrylate units, also show significant ion conductivity below T_g .

References

- [1] H.A. Every, F. Zhou, M. Forsyth, D.R. MacFarlane, *Electrochem. Acta*, in press.
- [2] T. Yamamoto, M. Inami, T. Kanbara, *Chem. Mater.* 6 (1994) 44.
- [3] T. Kanbara, M. Inami, T. Yamamoto, A. Nishikata, T. Tsuru, M. Watanabe, N. Ogata, *Chem. Lett.* (1989) 1913.
- [4] Forsyth et al., *ACS Symposium on Glassy Polymers*, April 1997, in press.
- [5] F. Zhou, Thesis, Monash University, 1996.
- [6] A. Ferry, *J. Chem. Phys.* 107 (1997) 9168.